Longlong Li

PhD Student | BioMEMS / Biosensors / Engineered Cardiac Tissue Monitoring

Department of Mechanical Engineering, Chonnam National University

Gwangju, South Korea | lilonglong2000@jnu.ac.kr | Homepage: https://longlongli.com

Condensed version for HR screening, R&D roles, and industry-facing applications.

Profile

- PhD student specializing in BioMEMS, multimodal biosensors, engineered cardiac tissue monitoring, and drug cardiotoxicity evaluation. Experienced in MEMS device design, microfabrication, cell and tissue culture, sensing system integration, and multimodal biosignal analysis. My research focuses on integrating strain sensors, microelectrode arrays, and 3D bioelectronic platforms for synchronized and quantitative assessment of cardiac tissue function.

Education

- 2022-Present | Ph.D. in Mechanical Engineering, Chonnam National University, Gwangju, South Korea
- 2018-2022 | B.Eng. in Mechanical Engineering, Wenzhou University, Wenzhou, China

Research Experience

- Research experience 1 | Designed and developed BioMEMS-based multimodal sensing platforms for engineered cardiac tissues.
- Research experience 2 | Integrated strain sensors and microelectrode arrays to monitor mechanical contraction and electrical activity in cardiac tissue models.
- Research experience 3 | Established 3D engineered heart tissue culture and drug stimulation workflows for cardiotoxicity screening and disease modeling.
- Research experience 4 | Developed MATLAB / Python-based analysis workflows for extracting contraction amplitude, beating frequency, field potential features, and electromechanical coupling parameters.
- Research experience 5 | Built experimental measurement systems involving low-noise signal acquisition, microscopy, sensor calibration, and hardware integration.

Technical Skills

- Microfabrication | Photolithography, metal deposition, SU-8 / PI processing, MEMS sensor fabrication, device packaging, sensor characterization.
- Biosensing | Strain sensors, microelectrode arrays, electromechanical monitoring, low-noise signal acquisition, sensor calibration.
- Cell & Tissue Engineering | NRVM / hiPSC-CMs culture, engineered heart tissue construction, collagen / dECM bioinks, drug stimulation experiments, immunostaining.
- Data Analysis | MATLAB, Python, cardiac signal processing, feature extraction, visualization, statistical analysis.
- System Integration | DAQ systems, amplifier circuits, microscopy, automated measurement platforms, prototype development.

Selected Projects

- 2026.03-Present | Bioinspired Hypoxic Cardiac Model and Electromechanical Evaluation Platform | MNTL | Developed an in vitro cardiac hypoxia model combined with multimodal electromechanical readouts. | Integrated mechanical contraction sensing and electrical activity recording for functional assessment of engineered cardiac tissues. | Applied the platform to evaluate cardiac tissue response under disease-relevant microenvironmental conditions.
- 2024.03-2026.03 | High-Throughput Drug Toxicity Screening Platform Based on 3D Cardiac Tissues | MNTL | Designed and optimized a 3D cardiac tissue-based sensing platform for drug cardiotoxicity evaluation. | Integrated strain-based mechanical sensing and electrophysiological readouts. | Supported long-term monitoring and multiparametric analysis of drug-induced functional responses.
- 2022.09-2024.03 | Graphene-Conductive Microenvironment for Cardiomyocyte Maturation and Electrophysiological Communication | MNTL | Investigated conductive microenvironment effects on cardiomyocyte maturation and intercellular communication. | Developed graphene/SU-8-based platforms for cardiac tissue functional enhancement. | Contributed to journal publication and conference outputs.

Selected Publications

- Air-Breakdown Triboelectric Nanogenerator Inspired by Transistor Architecture for Low-Force Human-Machine Interfaces. Karthikeyan Munirathinam, Longlong Li, Arunkumar Shanmugasundaram, Jongsung Park, Dong-Weon Lee. Nano-Micro Letters, 18(1), 251, 2026. Link: 10.1007/s40820-026-02103-0

- Harnessing native blueprints for designing bioinks to bioprint functional cardiac tissue. Mst Zobaida Akter, Fatima Tufail, Ashfaq Ahmad, Yoon Wha Oh, Jung Min Kim, Seoyeon Kim, Md Mehedee Hasan, Longlong Li, Dong-Weon Lee, Yong Sook Kim, Su-jin Lee, Hyung-Seok Kim, Youngkeun Ahn, Yeong-Jin Choi, Hee-Gyeong Yi. iScience, 28(3), 2025. Link: https://www.cell.com/iscience/fulltext/S2589-0042(25)00142-7
- Development of multifunctional PAA-alginate-carboxymethyl cellulose hydrogel-loaded fiber-reinforced biomimetic scaffolds for controlled release of curcumin. Kamrun Nahar Fatema, Longlong Li, Khurshid Ahmad, Jongyun Kim, Dong-Weon Lee. International Journal of Biological Macromolecules, 301, 140449, 2025. Link: 10.1016/j.ijbiomac.2025.140449
- Enhancing cardiomyocyte maturation through PEDOT:PSS-coated surfaces and mechanical stimulation with strain sensors. Jongyun Kim, Arunkumar Shanmugasundaram, Longlong Li, Zhuo Qu, Eung-Sam Kim, Bong-Kee Lee, Dong-Weon Lee. Journal of Micromechanics and Microengineering, 35(4), 045002, 2025. Link: 10.1088/1361-6439/adb8f9
- Graphene SU-8 platform for enhanced cardiomyocyte maturation and intercellular communication in cardiac drug screening. Longlong Li, Arunkumar Shanmugasundaram, Jongyun Kim, Nomin-Erdene Oyunbaatar, Pooja P. Kanade, Seong-Eung Cha, Daeyun Lim, Chil-Hyoung Lee, Young-Baek Kim, Bong-Kee Lee, Eung-Sam Kim, Dong-Weon Lee. ACS Nano, 18(49), 33293-33309, 2024. Link: 10.1021/acsnano.4c05365

Representative Conferences

- 2024 | Enhancing Cardiomyocyte Maturation via Mechanical Stimulation of 3D Printed Cardiac Tissue Using a Origami-based 3D Sensor and Magnetic Fields | IEEE NANOMED 2024, Hawaii, USA | IEEE NANOMED 2024, Hawaii, USA | Longlong Li, Haolan Sun, Jong-Yun Kim, Yun-Jin Jeong, Bong-Kee Lee, Eung-Sam Kim, Dong-Weon Lee
- 2025 | A Dual-Detection Approach for Cardiotoxicity Screening: Utilizing Nano Silicon Strain Sensor and Mea to Monitor Contractility and Field Potential in Cardiomyocytes | IEEE MEMS 2025, Kaohsiung, Taiwan | IEEE MEMS 2025, Kaohsiung, Taiwan | Haolan Sun, Longlong Li, Dong-Weon Lee
- 2025 | Origami-Inspired 3D Sensor Platform for Real-Time Electromechanical Coupling Analysis in Engineered Heart Tissues | IEEE SENSORS 2025, Vancouver, Canada | IEEE SENSORS 2025, Vancouver, Canada | Longlong Li, Fatima Tufail, Haolan Sun, Jongyun Kim, Hee-Gyeong Yi, Dong-Weon Lee
- 2025 | Development of a Multi-Channel Wireless Monitoring Platform for Long-Term Cardiomyocyte Contraction Assessment Using a Polymer Cantilever with integrated Sensor | MicroTAS 2025, Adelaide, Australia | MicroTAS 2025, Adelaide, Australia | Ke Liu, Haolan Sun, Longlong Li, Dong-Weon Lee
- 2026 | Multimodal Microelectrode-Microcantilever Array for Electromechanical Analysis of Cardiomyocyte Tissue in Drug Testing | IEEE MEMS 2026, Salzburg, Austria | IEEE MEMS 2026, Salzburg, Austria | Haolan Sun, Longlong Li, Eung-Sam Kim, Bong-Kee Lee, Dong-Weon Lee

Patents

- Patent Summary | 15 Total Patents | 4 Invention Patents | 3 Granted Inventions | 11 Utility Models
- Vertical-Folding-Based Clothes Folding Machine | Vertical-Folding-Based Clothes Folding Machine Granted Invention Patent, CN113186699B, 2023-02-07.
- Household Intelligent Toy Storage Robot | Household Intelligent Toy Storage Robot Granted Invention Patent, CN112407980B, 2022-01-18.
- Automatic Potato Harvester | Automatic Potato Harvester Utility Model Patent, CN216415123U, 2022-05-03.
- Intelligent Launch-and-Recovery Management Device for UAV Swarms | Intelligent Launch-and-Recovery Management Device for UAV Swarms Utility Model Patent, CN216269980U, 2022-04-12.

Awards

- 2025 | BK21 Fellowship Scholarship | Academic Fellowship
- 2025 | RLRC Outstanding Graduate Student | Graduate Recognition
- 2021 | China National Scholarship | National Scholarship
- 2019 | Zhejiang Provincial Government Scholarship | Provincial Scholarship